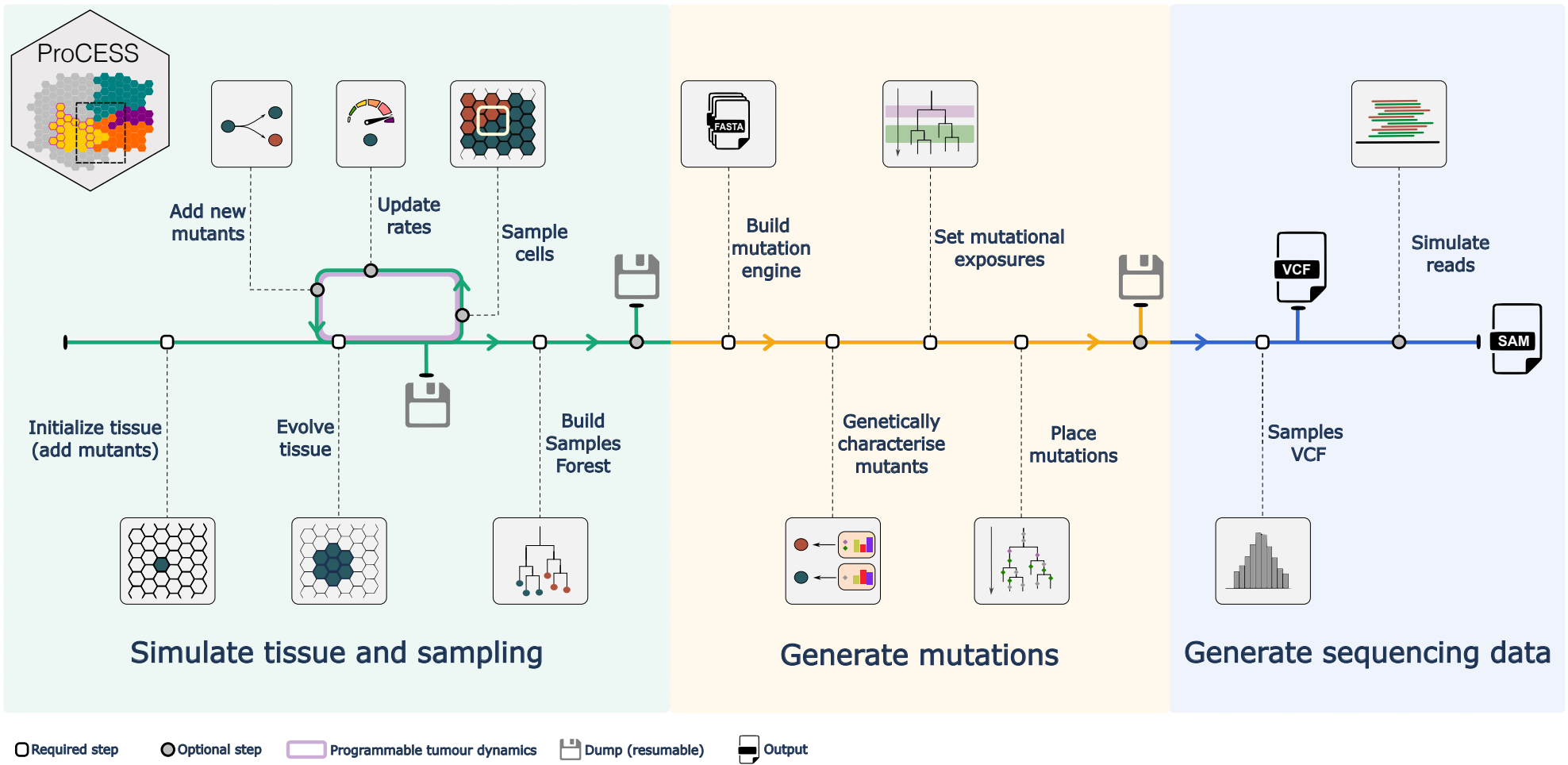




Advanced tissue-sampling strategies

The tissue can be sampled at single (a) or multiple time-points, and in multiple spatially-separated positions (b).

Single-cell phylogenies

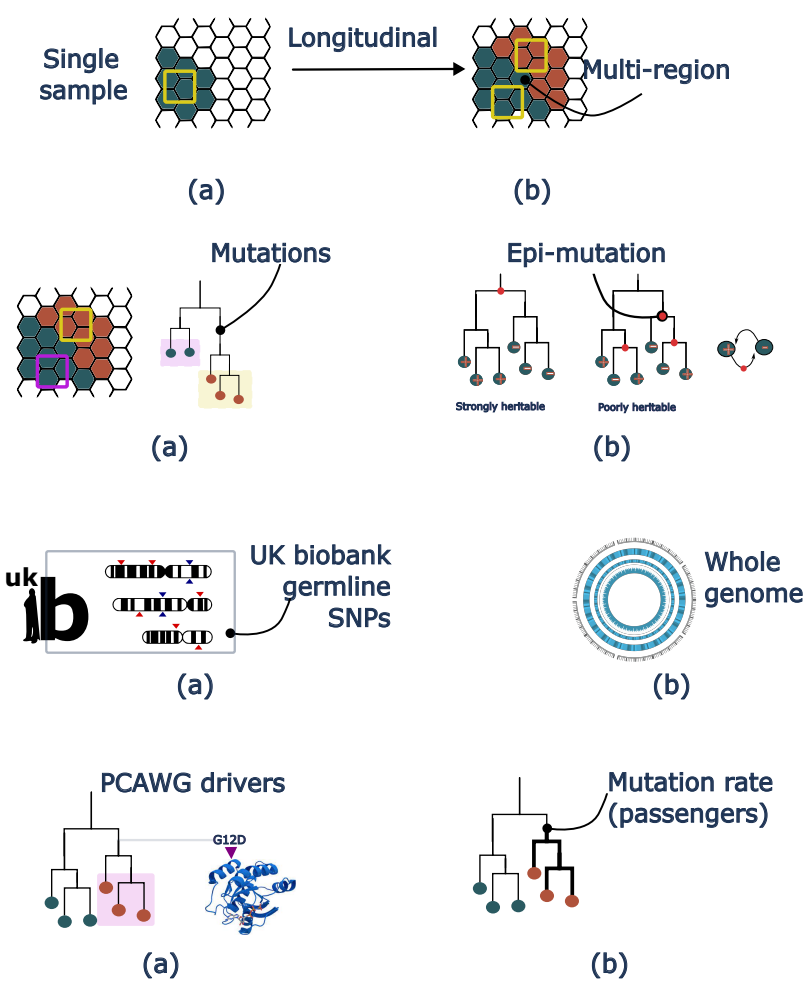
Single-cell phylogenies are generated from tissue samples (a), capturing epi-mutations with different heritability patterns (b).

Realistic mutational process

Germline reference from the UK biobank samples (a), with possibility of simulating whole-genome sequencing datasets (b).

Mutant genomic characterisation

Custom driver events per mutant (a) and passenger mutation rates per event type (b).



In lattice tissue simulation

A tissue is a squared lattice (a), with cells positioned in a discrete coordinate system (b), and pushing each other during simulation.

Modes of cell division on the tissue

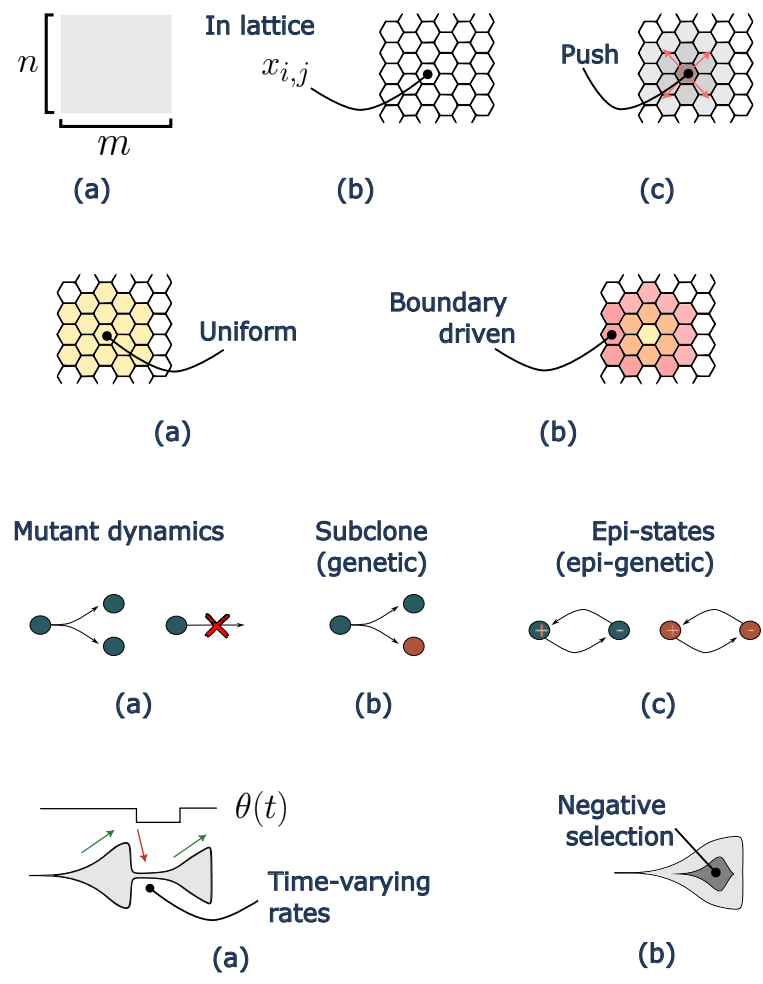
Tissue evolution is stochastic and depends on cell parameters. Divisions happen uniformly on the lattice (a) or on boundaries (b).

Advanced birth-death process

A stochastic birth-death process (a) with subclones with custom parameters (b), and reversible epi-genetic events (c).

Time-varying evolutionary parameters

The parameters of the birth-death process can vary in time to simulate therapy (a) effect and model negative selection (b).



Realistic mutational processes

Time-varying mutational exposures to emulate environment changes (e.g., therapy).

Phylogeny-consistent mutations

Somatic events are mapped on branches of the simulated phylogeny according to active mutational signatures.

VCF output

Mutation occurrence counts data in VCF format, NovaSeq error model (a), to create variant allele frequency data (b).

SAM output

SAM output can be streamlined with standard bioinformatics pipelines.

